CHAPTER 6

PROSPERITY

THE FINNISH NATIONAL epic *Kalevala* begins with Väinämöinen, the first man to bring trees and life to a previously barren world, now washed up on the shores of distant Pohjola after an exhausting defeat in a battle at sea.[1] After nursing the hero back to health, Louhi, the wicked queen who rules over Pohjola's dark and shadowy land, demands repayment in return for his release. Not content with mere gold or silver, the old hag of the north requests what then existed only in myth: the Sampo, a magical machine capable of producing for its owner an endless fountain of wealth.

With his brother Ilmarinen—architect of heaven's dome and the only person capable of creating anything like the Sampo—Väinämöinen strikes a deal with the powerful witch, promising that if she releases him, he will honor his debts by sending his brother back in his place. Louhi then seduces Ilmarinen with the prospect of receiving one of her beautiful maiden daughters as a bride, and the master craftsman enthusiastically does as Louhi commands.

Summoning divine winds to work the bellows for three days, the eternal hammerer forges the finest materials in the realm—"the tips of white swan feathers, the milk of greatest virtue, a single grain of barley, and the finest wool of lambskins"—into the Sampo, grafting spigots onto its sides that gush forth an infinite supply of grain, salt, and coin.[2] But just as Ilmarinen pulls the wealth-making machine from the forge's flames, Louhi gleefully snatches it out of his hands and locks it inside a mountain vault. Henceforth, Pohjola will prosper

from its limitless productive powers while Ilmarinen is left bitter and dejected.

Many years later, Ilmarinen and Väinämöinen return together to correct the witch's injustice. Upon arrival in the kingdom of plenty, they threaten to take the Sampo by force if they do not receive half of its profits. A fierce battle ensues at sea, and amid the chaos the Sampo sinks deep into the black depths where it remains, beyond retrieval, churning out riches and commodities without a master while rendering the ocean's water salty until this very day.

Similar stories describing machines of bounty exist throughout the world: the Akshaya Patra, a bottomless copper vessel described in the Hindu epic *Mahabharata*; the magical Cauldron of Plenty in the Irish myth of the god Dagda; the *Uchide-no-kozuchi*, a magic mallet in Japanese lore capable of "tapping out" anything upon command, including houses, clothing, and even humans.[3]

Those building AI today believe that their creation will constitute this fully stocked granary, this magic mill, this cornucopia overflowing with flowers, fruit, and corn. As mythology warns, however, creation alone will not be enough. To fulfill its potential, AI must, in its development and deployment, be aided by and coupled with suitable institutional changes and wise policy design. AI should be used to loosen, and ideally remove entirely, the bonds of servitude that in the past have dominated humanity's social and economic relations, and to launch us into a future characterized by less poverty and inequality.

This goal is extraordinarily ambitious by any measure, and one can rightly raise questions. Still, what if AI were indeed to offer a bridge to a new golden age? Even a partial success could amount to a civilizational renaissance.[4]

GROWTH AND INCLUSIVITY

In March 2016, after losses in three straight games, Lee Sedol, the Korean grand master of the Chinese game Go, felt neither anger nor

sadness but wonder. He never imagined that he—having dedicated his entire life to the mastery of this ancient game—could lose to AI, an ultramodern foe. And yet, only a game earlier, his computational opponent, known as AlphaGo, had played a move—number 37—so unorthodox as to force him to consider anew that machines might not only possess raw capability but also wield powers of creativity.

By now, awed by his extraordinary competitor, Lee was no longer playing solely in pursuit of victory—which he had already conceded in their best-of-five match—but was instead striving for a beautiful finish. In the next and fourth game, he responded to Move 37 with Move 78: a stroke of corresponding artistry that propelled him against all odds to victory in what is still the only game that AlphaGo has ever lost. In Korea and around the world, commiseration gave way, temporarily, to celebration.

Throughout that week in Seoul, the world's number one player, with only the occasional reprieve for a smoke on the outdoor patio of the sponsoring luxury hotel, had fought a lonely fight, locked in a contest with a unique adversary that he never expected to face while playing on behalf of a team—humanity—that he had never sought to represent. Their contest will be most remembered not for its final outcome but, all things considered, for the astounding capacity on display by human beings: namely, both Lee himself and those who developed his machine rival.[5]

The motto of DeepMind, the company behind the victory in Korea, is "First, solve intelligence; then, use intelligence to solve everything else."[6] Intelligence, as an engine of fresh creation, is set to change our understanding of everything. Facing this great unknown, we could become overwhelmed. But in at least some contexts, it may be wise to respond with the attitude shown by Lee Sedol, who in facing AI treated it as an inspiration rather than as a rival.

Lee Sedol's situation was unique—that of an expert at the top of his field, playing against an AI as an experiment. As a result, he may have been naturally suited to a posture of awe and imagination, as opposed to one of resentment. Many other people will respond to AI

much more negatively.

Refraining from rivalry will be particularly difficult in what appear to be zero-sum situations, perhaps especially AI's potential displacement of human labor. In this chapter we attempt to address that supposed zero-sum dynamic—which we see as largely misperceived—and to describe what we believe could be abundant and widely distributed benefits to humanity, even, potentially, in a world without work.

For most human beings throughout most of history, labor has not been a game one hopes to win or an art form one hopes to master but instead an unfulfilling and brutal burden, enforced through social structures that keep workers bound in service. Though these structures may have helped to maintain stability, they have invariably served also to torment the human spirit.

In the *Bhagavad Gita*, part of the *Mahabharata*, a dialogue between the warrior prince Arjuna and his charioteer—the deity Krishna in disguise—offers a pertinent discourse on the socioreligious hierarchies that long provided order to Indian society. As Arjuna hesitates on the battlefield to raise a sword against his own kin, Krishna explains to him in no ambiguous terms that there can be no deviation from duty, no divergence from destiny: "Better botch your job than gain perfection in your neighbor's; die if you must, but do not run the risk of alien labors."[7]

In this context, everyone plays a specific role, no matter how unfulfilling, as unfairly determined by birth. Thus, according to the *Gita*, "The duty of Brahmins [the highest caste] is to be peaceful and wise; of soldiers, to fight; of the middle classes, to care for farming and trade; and of serfs, to perform menial services."[8] Only if everyone faithfully discharges the duties of his or her station can a society succeed. Those who do so in this life have a chance at higher status in the next; for those who don't, suffering in the next reincarnation awaits.

Hindu castes, of course, were not alone. Aristotle's political theory involved strict social roles and duties. Slavery—underpinned by law, force, and psychological torment—became in parts of the world the

primary cruel institution for extracting labor and enforcing constructed social rank.

Over the last two centuries, capitalist democracies have largely replaced both caste and captivity with markets of meritocracy; churchmen have elevated and scholars have documented the societal values of a strong work ethic; and workers have acquired the arts of bargaining and strikes. But even so, and whether our human labors have been employed by the divine, or by government, or to guarantee a wage, our exertion of mind and body has generally been not so much for ourselves as in service to another.

Many a war has been caused by or else has resulted in changes to the question "Who gets what—and why?" (to borrow the title of the 2015 bestseller by the economist Alvin Roth). Earth's relatively fixed supply of land, labor, and capital has ensured that scarcity—not abundance—has been the predominant paradigm of economic theory and practice. Fierce battles are fought over how to divide what has been created, and even more often over how to distribute what little remains. These frictions pertain as much within societies as among them, even in peacetime, with citizens debating the roots of relative disadvantage and calling for redistribution to address the fact of widespread suffering.

An enlargement of the total amount of wealth available for redistribution, and subsequently an actually enlarged volume of redistributed wealth, would raise human standards of living around the world. If executed at enormous scale—the scale necessary to convince any given society, or any entity within a society, of the sufficiency of its wealth—such a development could transcend contemporary discussions of sustenance and focus our attention instead on abundance.

Enter AI—which presents a real opportunity to displace at least one among the original factors of production by shifting the function of labor from humans onto machines. Moreover, AI will be put to use researching and developing increasingly cheap and abundant sources of raw materials for its own inputs. As AI is simultaneously deployed in manufacturing, it could reduce the capital needed for any given good. True, a few nonrenewable elements and

commodities will continue to be needed to equip nonhuman intelligence itself, but that may change if AI is successfully deployed to find or generate synthetic substitutes. A new computing architecture, more efficient than today's by orders of magnitude, could be redesigned by AI, and eventually so could the factories that make AI's constituent components.

In producing more sustainable synthetic substitutes for a wide variety of goods, AI could enable a new age of abundance. Even given some ongoing physical and material constraints, its contributions—though hardly infinite—could be of such magnitude as to fulfill all of humanity's basic needs and realize many of our hopes. This may relax the grip that the paradigm of scarcity has had on our psychology, as well as the pessimism induced by our obligation to work as a means of survival.

Sam Altman, the chief executive of OpenAI, has analyzed economic systems along two variables: growth and inclusivity.[9] Many societies have been able to achieve one or the other, at least for a period of time; far fewer have been able consistently to maintain both. Of this relationship, Altman writes:

> Capitalism is a powerful engine of economic growth because it rewards people for investing in assets that generate value over time, which is an effective incentive system for creating and distributing technological gains. But the price of progress in capitalism is inequality.

In other words, AI and its attendant productivity gains may naturally catalyze a prolonged period of growth. But inclusivity will occur only by choice.

In a post-AI world, therefore, perhaps the solution would be, as Altman suggests, to tax the two "assets that will make up most of the value in [the] world," namely, companies—especially those that build, maintain, and use AI—and land, which still remains fixed (at least here on Earth). Certainly, if individual humans are not responsible for the value of labor that comes from AI-generated

insights, it is logical that this value be shared. And it is very possible that land (and, for a time, the rare-earth minerals essential to computing) will be among the few truly fixed—and thus valuable, and thus taxable—assets in a post-scarcity world.

But this suggestion assumes the continued existence of nations (or their recognizable substitutes) as redistributive agents, and of companies as potential targets for taxation. Moreover, placing a premium on both land and innovation could incite endless and possibly violent struggles over primacy. An alternative vision for ensuring equity in the age of AI might be found in a function analogous to the stock market: namely, the creation and automatic global assignation of divisible units of wealth associated with the increasing profits of AI models. (Monetary rights might come, as some shares do, with voting rights.)

Still another possibility would be to focus less on the ownership of AI than on the distribution of its ultimate benefits. But this would be subject to opposition, both on the obvious grounds that it should instead distribute ownership of the means of production, and on the practical grounds that it would require a huge amount of logistics and monitoring to ensure a particular standard across billions of individuals.

A further option could take inspiration from the patent system: allowing exclusive ownership of an AI invention and its profits so as to incentivize improvement, but only for a limited period; beyond a certain date, assuming its safety has been proven, the model could be published (or, perhaps, copies made and infrastructure built in new locales) for common use, iteration, and gain.

In brief: AI pioneers may underestimate the scope of the economic and political challenges that they have set in motion. Yes, AI will be able to do just about anything. But, as Sam Altman puts it bluntly, "Is it going to do what *I* want, or is it going to do what *you* want?"[10] How will "we" decide, and who is "we"? To direct these energies of enormous possibility, and to redistribute the benefits of those directions, is a grave responsibility. Future decision-makers must take care not to entrench once again the sorts of social and

economic inequalities that spread outward from the Industrial Revolution before beginning to be corrected, much too slowly, through more human-directed structures of control.[11] [12]

Today the preponderance of benefits of, and almost exclusive control over, advanced AIs has accrued to extremely few individuals. Will they give up their advantages? If they do—if and when more benefits and more control come to be shared at the national level—then calls for the globalization of both will begin immediately. Will one nation turn its sovereign wealth into common benefit? Some may argue that the psychological barriers to sacrifice will disappear once the world is no longer zero-sum. But this presupposes a transition that has not yet come to pass, one that seems contrary to today's status quo, and one that—if it is to pass—would need to be the product of human choice.

Moreover, even if we manage to arrive in a world beyond scarcity, it is not clear how global human incentives would be structured to remain in harmony. A world beyond value is not beyond values. Those who do not commune on the basis of money may commune—and then proceed to rearrange society and global institutions—on the basis of religion, race, family lineage, education, morality, skill, aesthetics, humor, or any other category. Those who do not fight for money may fight for God, power, glory, or revenge.

Furthermore, economic history confirms the difficulty of devising systems as consistent as they are effective. And one must admit that humans have been inept at predicting the long-term effects of technology; in the case of AI, our optimism may be unjustified and our apprehensions misplaced.

Nevertheless, the authors of this book believe that AI could conceivably be harnessed to generate a new baseline of human wealth and well-being—and just that possibility itself demands that we make a start in this direction. Moreover, we are confident that if such an economic and political scheme comes to pass, it would at least ease if not eliminate the strains of labor, class, and conflict that previously have torn humanity apart.

MOBILITY

If, in the age of AI, our cup runneth over, how can we ensure that every human is capable of remotely and reliably benefiting from the new surplus? Already, humanity has made great strides in the distribution of value: from the earliest currencies, which lightened the burdens formerly imposed by systems of barter, to later iterations of fiat currencies in paper and coin, as well as in such digital inventions as credit cards, wire transfers, and mobile banking. While these vessels have facilitated a more efficient movement of value across space, due to their jittery supply they have underperformed in their value over time. Modern projects in currency seek the efficient transfer of value across both space and time.

The modern economy is still formed by the production of goods and the provision of services, not by their indirect, abstract representations. Money is null without a marketplace; it becomes only an entry in a database for resource allocation, in which value ceases to hold meaning. Viewed through the lens of information theory, money today has much in common with an internet connection, which for any recognizable use similarly requires relational context. How best to optimize its technical attributes may in the age of AI become a task of immense philosophical and technical urgency. Quite possibly, the world will need a new type of financial network that balances the traditional functions of money as a consistent store of value and convenient means of exchange.

AI could also introduce new dynamics to financial markets and economic policies. It is easy to imagine, abstractly and in principle, how AI might go about creating wealth—even with access only to the internet. But even if we can invent currencies, systems, markets, and policies that respond wisely to the emergence of this scale of potential value-generation, how, exactly, would AI solve or end poverty, for example? As a practical matter, how would it establish an absolute global baseline for our quality of life?

If AI proves to be capable of physically distributing the goods that

humans require for our fundamental material needs, then the volume of materials that could be generated and moved around the world will be unprecedented. There are undoubtedly many possibilities for such a venture—and just as many pitfalls. Distributed AI systems might be designed to connect, via mass-manufactured robots and AI-optimized infrastructural systems, to the unconnected. Those new connections—lacking for some 2.6 billion people today—would enable the provision of food, clothing, and shelter to the half of humanity that still lacks those basic necessities.[13] Alternatively, using new sustainable synthetic materials, AIs could build cities around the world to provide shelter, regulate temperature, ensure access to power and digital connections, and provide clean water, food, medicine, and sanitation. Machine intelligences of various complexity could even service these cities, which could house tens of millions of people who might otherwise be living outside the system of prosperity that too few of us enjoy today.

Perhaps both alternatives would be realized, alongside other futures that we have not yet imagined. In that case, humans would retain their options—and likely enjoy more freedom of choice than before. With humans no longer bound to the geography of their birth, the communities of their kin, or the labor markets for their skill, where would they move if immigration were transformed from a tragic necessity for some to a free choice for all? If a large number of new cities of the same high quality were to be built by AI, would we still see mass migration from undeveloped to developed nations, or from rural settings to urban ones, as historically has been the case?

Greater psychological freedom in the formation of families might also become a reality thanks to AI. Some might choose to have more children, freed of the necessity to make damning decisions about how to allocate resources among multiple children based on which of them have the greatest chance of survival and success. Others, if no longer reliant on offspring for support in old age, might refrain from procreation. Generational burdens could dissipate, enabling children to pursue geographies and professions they might

previously have discounted.

Earlier imbalances and asymmetries in the global economy have derived from differences in, among other factors, resource endowment, geography, and human capital. AI could reduce talent gaps and equalize resource distributions. That would render less relevant the predominant fault lines on which we have long conducted global trade and commerce and by which the world's prosperity is stratified. Countries that today are disadvantaged by their territorial or other endowments, or that are suffering from "brain drain," could find themselves with new means of advancing to the same circumstantial standards as those enjoyed by the globe's traditional economic giants.

How do we transition from today's international inequality to the future we describe? The first step, perhaps, is the design of AI-enabled systems and applications to advance material sciences and optimize digital connections—including the purpose-built datasets that would enable such systems to function in a variety of global contexts. As we invest in these ideas, we should keep in mind the potential gains: a profound transformation in the overall standard of collective human life, and in the equality of individual lives across race, gender, nationality, place of birth, and family background. An equalization of the cost of intelligence, distributed across the globe, could bring about a level playing field that has never before existed.

ABUNDANCE WITHOUT ABANDON

But what if AI, while acting as an economic equalizer, sends the cost of intelligence, and therefore of labor, plummeting toward zero? That would conclude the brief but wonderfully productive period of human history that has allowed individuals in free societies to improve their circumstances, should they so choose, by their own efforts. As scarcity has been the paradigm of the past, competition—at least in the modern era—has been the default condition of self-organization, leading naturally to wide variations in the distribution

of outcomes as a function of ambition, ability, and the lottery of where and to whom one was born.

All of this has meant that, by and large, humans more industrious in the application of their labors or in the harnessing of others' labors have fared better while others have fared worse. If, however, we remove the sorting function of labor, we must also contend with the removal of professions and the status, identity, and meaning that are associated with them. This would be a different world indeed.

Our natural instincts for overcoming adversity, for celebrating excellence, for taking pride in notable difference and diversity, will surely still exist, though they will need to seek new channels. Just as once there were distributions of uneven human talents for labor, there may arise new distributions of uneven human talents for leisure. And, this time, such distributions might fall not along existing axes of ability but rather along axes of a different quality: curiosity, sobriety, kindness, or perhaps something else entirely.

In a world without work, many of us might become entranced and engulfed by simulated, custom-immersive worlds: a sensuous orchestra of sight, sound, smell, touch, and even taste now made possible by AI's complete powers in the virtual realm. As discussed in Chapter 3, billions of passive humans could plausibly choose—or be lured down—this path, finding it neither easy nor necessary to resist so instantaneous an upgrade in the level of stimulation in, and the feeling of control over, their reality.

Even today, when people are not otherwise occupied with the tribulations of life and labor, the light in many eyes is likely to reflect the glow of pixeled plates of glass. In every corner of the world, workers, exhausted from their days of toil, understandably retreat to this lighter, purer medium to consume and to create.[14] If humans already struggle to restrain their attraction to today's relatively primitive technologies, how will we ever handle the vastly superior "experience machines" that are sure to be enabled by AI?[15] If many of us can barely tolerate the pain of work, how will we resist unqualified pleasure?

The answer is that human psychology will need to coevolve with

AI and its effects. It is difficult to predict how, precisely, but it seems possible that AI could in fact enable human meaning as much as, or more than, it detracts from it. Pleasure alone will not satisfy our innate desire for meaning. Work—even if not for pay—can provide a sense of purpose insofar as the means are strenuous or the ends are noble.

A personal experience of difficulty can instill a sense of pride once that challenge is overcome. Strenuous toil, especially when combined with commitment, lends its own narrative arc to our understanding of time, the self, and the human capacity for mastery. Given our human psychology, many sources of joy and contentment may well remain unchanged in the age of AI, despite the myriad changes that are sure to sweep through our lives.

This is not to say that we will simply revert to activities of the past; instead, we may discover aspects of human potential that we had not previously pursued at scale. Consider the possibilities that might emerge from periods of human focus that until now have been impossible to include in an ordinary person's working week. Mental and spiritual exercises, performed at length, could elevate human consciousness. Extended periods of heightened awareness may in turn assist our relational connections to other humans (and animals), strengthen our perceptions of the divine, and produce meaningfully elevated levels of individual well-being.

Unassisted humans performing apparently superhuman feats, especially those involving the use of our physical bodies, will no doubt remain events of fascination. With greater numbers of humans entering and attempting to master such activities, one might expect soaring excellence. Sports and games that push the human to the limit could increase in prevalence and quality. Art could flourish, for the ring of the authentic is likely to retain its charm.

Universities historically committed themselves to giving their students an equal introduction both to the sciences and to the humanities.[16] We believe these quintessentially human endeavors, each in its own way a "quest for meaning" (to borrow a phrase from one spiritual leader),[17] will if anything expand. In the West, the

liberal arts were the subjects and skills held by classical antiquity to be markers of an independent mind. In the ancient Sinosphere, the Tang dynasty artist Zhang Yanyuan established that a "scholar-gentleman" should be expected to demonstrate proficiency in "the four arts": abilities of sound, sight, strategy, and script.[18] In the future, and in contrast to schools of education that are vocational, professional, or technical, we may hope for a revival of some of the earliest attempts to nurture the "learned individual" capable and desirous of diverse calling. Schools around the world could produce the philosophers and writers we need for reorientation to an entirely new age.

Subjects once reserved for the privileged few could become standard for the many, replacing the average classroom's previous focus on the assembly of productive labor. Deployment of AI educators could enable individualized instruction and Socratic seminars around the world. Imagine if, just as a young Albert Einstein was tutored by Max Talmud (later Max Talmey), Voltaire by the Abbé de Châteauneuf, and Ada Lovelace (who wrote the first computer algorithm) by Mary Somerville, every child would now be fairly enabled in the same ways to master his or her mind and character.[19]

We can imagine that where universities stand today—true to their original form, an assemblage of dormitories encircling a library of books around which eager minds gather to attain and extend the intellectual frontier—

campuses might someday include spaces of congregation for teams of humans to interpret the discoveries of AI itself: that is, to understand them and to translate the most prominent among them into relevance for human life.

In this new branch of the sciences, humans might choose to coevolve with AI in order to remain partners with the machines at the frontier. Or that may turn out not to be necessary in order for us to maximize the benefits to humanity from AI's discoveries. In either case, and especially the latter, we expect this to be an extraordinarily difficult activity, working side by side with our machines in around-

the-clock shifts. But however grueling, the effort, further explored in Chapter 8, would be essential.

THE PRIVILEGE OF CHOICE

Faced with the perceived threat from the automation of human labor, many commentators today fixate on the advent of a new spiritual crisis: In a world of shared abundance, they argue, we would become like irresponsible lottery winners, overcome by the hedonism of excess. This, in our view, is a perspective of privilege. To appreciate the extraordinary good that AI will do for billions of people—including those who currently lack the money, connectivity, basic necessities, and leisure time to participate in ongoing conversations of our kind here—one need only call to mind the contrasting lot of their forebears: generations of fathers operating machines of primitive make, mothers struggling in fields of arid harvest, child laborers robbed of their innocence. If, tomorrow morning, every human were given the choice to stop working, we suspect most would do so, while the few who declined would likely be privileged to work not out of obligation but out of choice. AI could be directed to do what we no longer *have* to do, precisely so that we can do the things we *want* to do.

As detailed in Chapter 3, we do worry that a great fraction of humans could become primarily passive consumers of AI-generated content. But this is a concern that stems from the human tendency toward easy consumption—and, more troubling still, from what we can only assume will be AI's future perception of us. In other words, our concern about human passivity is not about the human loss of paid work. We already have a prototype of how people live when they can have what they want without working. We call them the rich and the retired. Certainly, the rich—including many who were not rich previously—are sometimes at a loss for how to exercise their options after so many years spent amassing them. As Tolstoy professed, "If a fairy had come and offered to fulfill my every wish, I

would not have known what to wish for."[20]

The adjustment to abundance is likely a problem of transition rather than a permanent challenge. Some will initially perceive the introduction of machine labors as depriving them of their primary source of fulfillment and joy. No doubt this will be a jarring experience. But to us it seems likely—not as a response to our exhortation but rather as an outgrowth of human instinct—that, given time, humans would choose to persevere, perhaps in new avenues or as partners of AI, avoiding atrophy and instead excelling as thinkers and doers. Ultimately, if we establish the needed systems for distribution, connection, participation, and education, humans—empowered and inspired by AI—may continue working not for pay but for pleasure and pride.